

Setup

Table 1 Simulation parameters

Basic parameters	3GPP TS 38.101 Subclause 5.2.4.1.1 Test 5-1
Modified parameters	No precoding, same information bit payload and binary channel bits for every slot
Data type	Floating point
Channel estimation, synchronization and noise power estimate	Perfect
Equalizer type	LMMSE equalizer provided by the platform
LDPC decoder	Layered belief propagation with 12 iterations, $N_{cb}=25344$

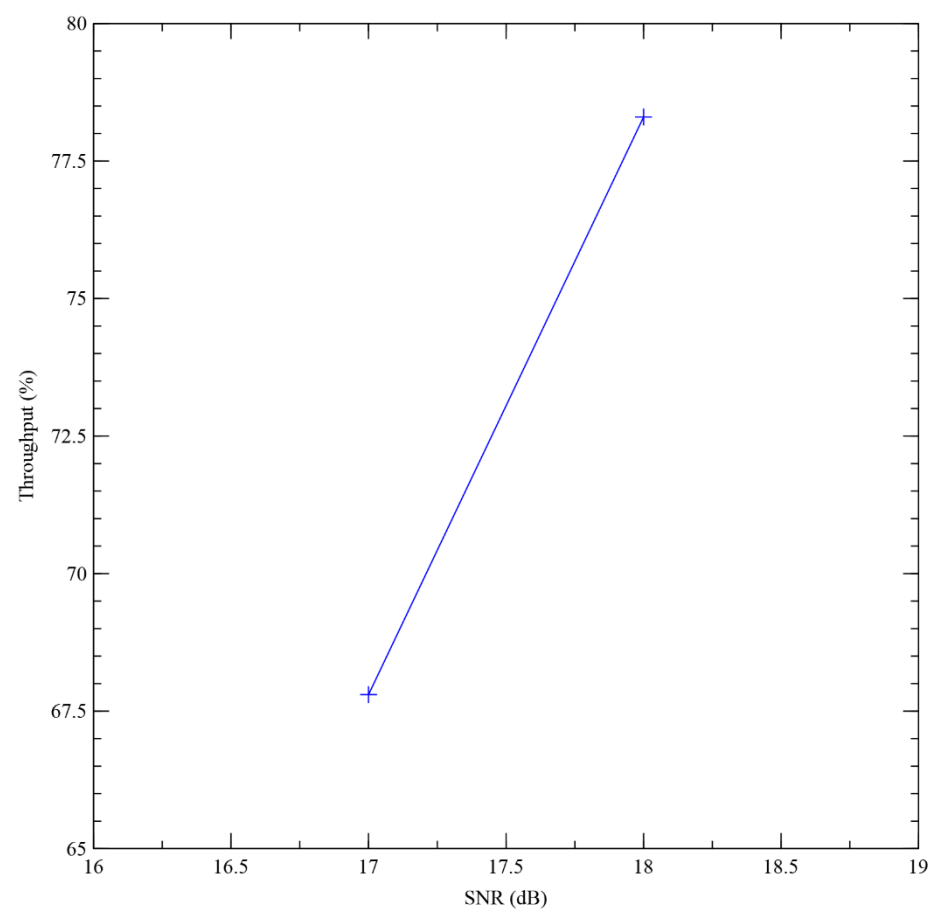


fig 1. LMMSE performance for 3GPP 38.101 5.2.4.1.1 test 5-1

Enhanced Receiver

Table 1 Simulation parameters

Basic parameters	3GPP TS 38.101 Subclause 5.2.4.1.1 Test 5-1
Modified parameters	No precoding, same information bit payload and binary channel bits for every slot
Data type	Floating point
Channel estimation, synchronization and noise power estimate	Perfect
Equalizer type	Enhanced receiver
LDPC decoder	Layered belief propagation with 12 iterations, $N_{cb}=25344$

Table 2 Gain of required SNR (dB) at 70% of the maximum throughput over conventional low complexity MIMO decoder

Modulation schemes and correlation matrix configurations	Low	Medium-C	Simulation duration (10 ms)
QPSK	0.x	1.x	100
16QAM	1.x	1.x	50
64QAM	1.x	1.x	10
256QAM	1.x	1.x	10
Note	-		